复旦数学资料汇编
数值代数及其应用

来源：香蕉空间 Solution 资料库

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1.1　试卷: 数值代数及其应用

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1. Answer the following questions briefly.
   - $A \in \mathbb{R}^{m\times n}$, $B \in \mathbb{R}^{n\times p}$, $C \in \mathbb{R}^{p\times q}$. How to compute $D = ABC$?
   - X, Y are 3×2 matrices, and $X^\top X = Y^\top Y = I_2$. What is the angle of these two spaces spanned by X, Y?
   - Compare **CGS, MGS, Givens, Householder** briefly when using them to compute QR factorization.
   - What symmetry eigenvalue methods can not be applied to the non–symmetry case?
   - Forgot :)
2. If A is column full rank, then prove:

$$\begin{pmatrix} I & A^\top \\ A & 0 \end{pmatrix} \begin{pmatrix} r \\ x \end{pmatrix} = \begin{pmatrix} b \\ 0 \end{pmatrix}$$

   is equivalent to the least square problem of (A, b).
3. If A is a strictly column dominate matrix, i.e., $\sum_{k\neq i} |A_{ki}| < |A_{ii}|$, then
   - Please prove that in **gepp** to compute the LU factorization of A, row exchanges never happen.
   - Please give an upper bound of the condition number of L; you can choose any norm you like.
5. Describe and prove the **Implicit Q Theorem**.
6. If A is an upper–triangular matrix with $A_{ii} \neq A_{jj}$ for all $i \neq j$, then please propose a pseudo–code to compute the eigenvectors of A, i.e., compute X, Λ such that

$$A = X\Lambda X^{-1}.$$

7. In **Divide & Conquer** method to solve the eigenvalue problem of symmetric matrices, w.l.o.g., we assume that

$$D + uu^\top$$

   - Elements in diag matrix D are different from each other,
   - $u_i \neq 0$.

   Please prove why this does not lose any generality.